

Robotics and Programming

Assignment 3 (100 pts)

Submission:

1. Submit a Python file (**LastName_studentID_hw3.py**) with your source code.
2. Do not compress your files.
3. If there is any plagiarism, you will lose all points on the questions the first time. Next, you will lose all points in the whole homework.
4. You can submit your homework **3 times** before the deadline.

Questions:

1. Low pass filter of an image. A low pass filter smooths an image. An original image is provided.



Figure 1: Original image

We use three different box filters (3x3, 5x5, and 7x7) to smooth the image. The template of the Python code is “Low_pass_filter.py”. You can work on it. You can add new code, **but do not change the template**.

The low pass filter will average the values in the box. The average value will be the new value in the center of the box. The resulting image consists of the average values. In the Python code, the operation of averaging the values in the box is implemented by multiplying a matrix of the low pass filter on the image in the **conv** function.

For example, the matrix of 3x3 box low pass filter is

$$filter = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \end{bmatrix}$$

We select a 3x3 matrix in an image,

$$subImage1 = \begin{bmatrix} 21 & 78 & 63 \\ 16 & 109 & 73 \\ 6 & 88 & 86 \end{bmatrix}$$

If we apply the filter on the sub-image, we will have.

$$C = filter * subImage1 = \begin{bmatrix} \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \end{bmatrix} * \begin{bmatrix} 21 & 78 & 63 \\ 16 & 109 & 73 \\ 6 & 88 & 86 \end{bmatrix} = \begin{bmatrix} 2.33 & 8.67 & 7 \\ 1.78 & 12.11 & 8.11 \\ 0.67 & 9.78 & 9.56 \end{bmatrix}$$

If we sum the values in the above matrix C , we will get 60. If we average the values in the above matrix $subImage1$, we will get 60 as well. So, we can see that the $filter$ matrix can do the low-pass filtering.

For 5x5 and 7x7 box low-pass filters, we can use the above method to create the filters. The result is shown below.

